

Group B Deep Learning

Assignment No: 1

Title of the Assignment: Linear regression by using Deep Neural network: Implement Boston housing price.prediction problem by Linear regression using Deep Neural network. Use Boston House price prediction dataset.

Objective of the Assignment: Students should be able to perform Linear regression by using Deep Neural network on Boston House Dataset.

Prerequisite:

1. Basic of programming language
 2. Concept of Linear Regression
 3. Concept of Deep Neural Network
-

Contents for Theory:

1. What is Linear Regression
 2. Example of Linear Regression
 3. Concept of Deep Neural Network
 4. How Deep Neural Network Work
 5. Code Explanation with Output
-

What is Linear Regression?

Linear regression is a statistical approach that is commonly used to model the relationship between a dependent variable and one or more independent variables. It assumes a linear relationship between the variables and uses mathematical methods to estimate the coefficients that best fit the data.

Deep neural networks are a type of machine learning algorithm that are modeled after the structure and function of the human brain. They consist of multiple layers of interconnected neurons that process data and learn from it to make predictions or classifications.

Linear regression using deep neural networks combines the principles of linear regression with the power of deep learning algorithms. In this approach, the input features are passed through one or more layers of neurons to extract features and then a linear regression model is applied to the output of the last layer to make predictions. The weights and biases of the neural network are adjusted during training to optimize the performance of the model.

This approach can be used for a variety of tasks, including predicting numerical values, such as stock prices or housing prices, and classifying data into categories, such as detecting whether an image contains a particular object or not. It is often used in fields such as finance, healthcare, and image recognition.

Example Of Linear Regression

A suitable example of linear regression using deep neural network would be predicting the price of a house based on various features such as the size of the house, the number of bedrooms, the location, and the age of the house.

In this example, the input features would be fed into a deep neural network, consisting of multiple layers of interconnected neurons. The first few layers of the network would learn to extract features from the input data, such as identifying patterns and correlations between the input features.

The output of the last layer would then be passed through a linear regression model, which would use the learned features to predict the price of the house.

During training, the weights and biases of the neural network would be adjusted to minimize the difference between the predicted price and the actual price of the house. This process is known as gradient descent, and it involves iteratively adjusting the model's parameters until the optimal values are reached.

Once the model is trained, it can be used to predict the price of a new house based on its features. This approach can be used in the real estate industry to provide accurate and reliable estimates of house prices, which can help both buyers and sellers make informed decisions.

Concept of Deep Neural Network-

A deep neural network is a type of machine learning algorithm that is modeled after the structure and function of the human brain. It consists of multiple layers of interconnected nodes, or artificial neurons, that process data and learn from it to make predictions or classifications.

Each layer of the network performs a specific type of processing on the data, such as identifying patterns or correlations between features, and passes the results to the next layer. The layers closest to the input are known as the "input layer", while the layers closest to the output are known as the "output layer".

The intermediate layers between the input and output layers are known as "hidden layers". These layers are responsible for extracting increasingly complex features from the input data, and can be deep (i.e., containing many hidden layers) or shallow (i.e., containing only a few hidden layers).

Deep neural networks are trained using a process known as backpropagation, which involves adjusting the weights and biases of the nodes based on the error between the predicted output and the actual output. This process is repeated for multiple iterations until the model reaches an optimal level of accuracy.

Deep neural networks are used in a variety of applications, such as image and speech recognition, natural language processing, and recommendation systems. They are capable of learning from vast amounts of data and can automatically extract features from raw data, making them a powerful tool for solving complex problems in a wide range of domains.

How Deep Neural Network Work-

Boston House Price Prediction is a common example used to illustrate how a deep neural network can work for regression tasks. The goal of this task is to predict the price of a house in Boston based on various features such as the number of rooms, crime rate, and accessibility to public transportation.

Here's how a deep neural network can work for Boston House Price Prediction:

1. **Data preprocessing:** The first step is to preprocess the data. This involves normalizing the input features to have a mean of 0 and a standard deviation of 1, which helps the network learn more efficiently. The dataset is then split into training and testing sets.
2. **Model architecture:** A deep neural network is then defined with multiple layers. The first layer is the input layer, which takes in the normalized features. This is followed by several hidden layers, which can be deep or shallow. The last layer is the output layer, which predicts the house price.
3. **Model training:** The model is then trained using the training set. During training, the weights and biases of the nodes are adjusted based on the error between the predicted output and the actual output. This is done using an optimization algorithm such as stochastic gradient descent.
4. **Model evaluation:** Once the model is trained, it is evaluated using the testing set. The

performance of the model is measured using metrics such as mean squared error or mean absolute error.

5. **Model prediction:** Finally, the trained model can be used to make predictions on new data, such as predicting the price of a new house in Boston based on its features.
6. By using a deep neural network for Boston House Price Prediction, we can obtain accurate predictions based on a large set of input features. This approach is scalable and can be used for other regression tasks as well.

Boston House Price Prediction Dataset-

Boston House Price Prediction is a well-known dataset in machine learning and is often used to demonstrate regression analysis techniques. The dataset contains information about 506 houses in Boston, Massachusetts, USA. The goal is to predict the median value of owner-occupied homes in thousands of dollars.

The dataset includes 13 input features, which are:

CRIM: per capita crime rate by town

ZN: proportion of residential land zoned for lots over 25,000 sq.ft.

INDUS: proportion of non-retail business acres per town

CHAS: Charles River dummy variable (1 if tract bounds river; 0 otherwise)

NOX: nitric oxides concentration (parts per 10 million)

RM: average number of rooms per dwelling

AGE: proportion of owner-occupied units built prior to 1940

DIS: weighted distances to five Boston employment centers

RAD: index of accessibility to radial highways

TAX: full-value property-tax rate per \$10,000

PTRATIO: pupil-teacher ratio by town

B: $1000(B_k - 0.63)^2$ where B_k is the proportion of black people by town

LSTAT: % lower status of the population

The output variable is the median value of owner-occupied homes in thousands of dollars (MEDV).

To predict the median value of owner-occupied homes, a regression model is trained on the dataset. The model can be a simple linear regression model or a more complex model, such as a deep neural network.

After the model is trained, it can be used to predict the median value of owner-occupied homes based on the input features. The model's accuracy can be evaluated using metrics such as mean squared error or

mean absolute error.

Boston House Price Prediction is an example of regression analysis and is often used to teach machine learning concepts. The dataset is also used in research to compare the performance of different regression models.

Source Code with Explanation-

```
#Importing the pandas for data processing and numpy for numerical
computing
import numpy as np
import pandas as pd
# Importing the Boston Housing dataset from the sklearn
from sklearn.datasets import load_boston
boston = load_boston()
#Converting the data into pandas dataframe
data = pd.DataFrame(boston.data)
#First look at the data
data.head()

      0      1      2      3      4      5      6      7      8      9     10     11     12
0  0.00632  18.0  2.31  0.0  0.538  6.575  65.2  4.0900  1.0  296.0  15.3  396.90  4.98
1  0.02731   0.0  7.07  0.0  0.469  6.421  78.9  4.9671  2.0  242.0  17.8  396.90  9.14
2  0.02729   0.0  7.07  0.0  0.469  7.185  61.1  4.9671  2.0  242.0  17.8  392.83  4.03
3  0.03237   0.0  2.18  0.0  0.458  6.998  45.8  6.0622  3.0  222.0  18.7  394.63  2.94
4  0.06905   0.0  2.18  0.0  0.458  7.147  54.2  6.0622  3.0  222.0  18.7  396.90  5.33

#Adding the feature names to the dataframe
data.columns = boston.feature_names
#Adding the target variable to the dataset
data['PRICE'] = boston.target
#Looking at the data with names and target variable
data.head(n=10)
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	PRICE
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1.0	296.0	15.3	396.90	4.98	24.0
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2.0	242.0	17.8	396.90	9.14	21.6
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2.0	242.0	17.8	392.83	4.03	34.7
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3.0	222.0	18.7	394.63	2.94	33.4
4	0.06905	0.0	2.18	0.0	0.458	7.147	54.2	6.0622	3.0	222.0	18.7	396.90	5.33	36.2
5	0.02985	0.0	2.18	0.0	0.458	6.430	58.7	6.0622	3.0	222.0	18.7	394.12	5.21	28.7
6	0.08829	12.5	7.87	0.0	0.524	6.012	66.6	5.5605	5.0	311.0	15.2	395.60	12.43	22.9
7	0.14455	12.5	7.87	0.0	0.524	6.172	96.1	5.9505	5.0	311.0	15.2	396.90	19.15	27.1
8	0.21124	12.5	7.87	0.0	0.524	5.631	100.0	6.0821	5.0	311.0	15.2	386.63	29.93	16.5
9	0.17004	12.5	7.87	0.0	0.524	6.004	85.9	6.5921	5.0	311.0	15.2	386.71	17.10	18.9

```
#Shape of the data
```

```
print(data.shape)
```

```
#Checking the null values in the dataset
```

```
data.isnull().sum()
```

```
CRIM      0
```

```
ZN        0
```

```
INDUS     0
```

```
CHAS      0
```

```
NOX       0
```

```
RM        0
```

```
AGE       0
```

```
DIS       0
```

```
RAD       0
```

```
TAX       0
```

```
PTRATIO   0
```

```
B         0
```

```
LSTAT     0
```

```
PRICE     0
```

```
dtype: int64
```

```
#Checking the statistics of the data
```

```
data.describe()
```

```
# This is sometimes very useful, for example if you look at the CRIM the max is 88.97 and 75% of the value is below 3.677083 and
```

```
# mean is 3.613524 so it means the max values is actually an outlier or there are
```

outliers present in the column

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO
count	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000
mean	3.613524	11.363636	11.136779	0.069170	0.554695	6.284634	68.574901	3.795043	9.549407	408.237154	18.455534
std	8.601545	23.322453	6.860353	0.253994	0.115878	0.702617	28.148861	2.105710	8.707259	168.537116	2.164946
min	0.006320	0.000000	0.460000	0.000000	0.385000	3.561000	2.900000	1.129600	1.000000	187.000000	12.600000
25%	0.082045	0.000000	5.190000	0.000000	0.449000	5.885500	45.025000	2.100175	4.000000	279.000000	17.400000
50%	0.256510	0.000000	9.690000	0.000000	0.538000	6.208500	77.500000	3.207450	5.000000	330.000000	19.050000
75%	3.677083	12.500000	18.100000	0.000000	0.624000	6.623500	94.075000	5.188425	24.000000	666.000000	20.200000
max	88.976200	100.000000	27.740000	1.000000	0.871000	8.780000	100.000000	12.126500	24.000000	711.000000	22.000000

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 506 entries, 0 to 505
```

```
Data columns (total 14 columns):
```

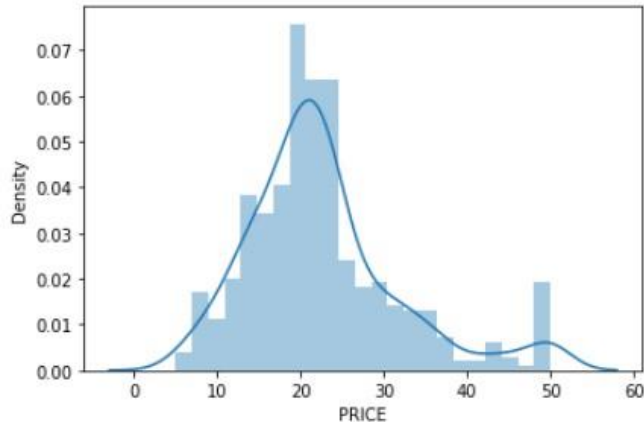
#	Column	Non-Null Count	Dtype
0	CRIM	506 non-null	float64
1	ZN	506 non-null	float64
2	INDUS	506 non-null	float64
3	CHAS	506 non-null	float64
4	NOX	506 non-null	float64
5	RM	506 non-null	float64
6	AGE	506 non-null	float64
7	DIS	506 non-null	float64
8	RAD	506 non-null	float64
9	TAX	506 non-null	float64
10	PTRATIO	506 non-null	float64
11	B	506 non-null	float64
12	LSTAT	506 non-null	float64
13	PRICE	506 non-null	float64

```
dtypes: float64(14)
```

```
memory usage: 55.5 KB
```

```
#checking the distribution of the target variable
import seaborn as sns
sns.distplot(data.PRICE)
#The distribution seems normal, has not be the data normal we would have perform
log transformation or took to square root of the data to make the data normal.
# Normal distribution is need for the machine learning for better predictiblity
of the model
```

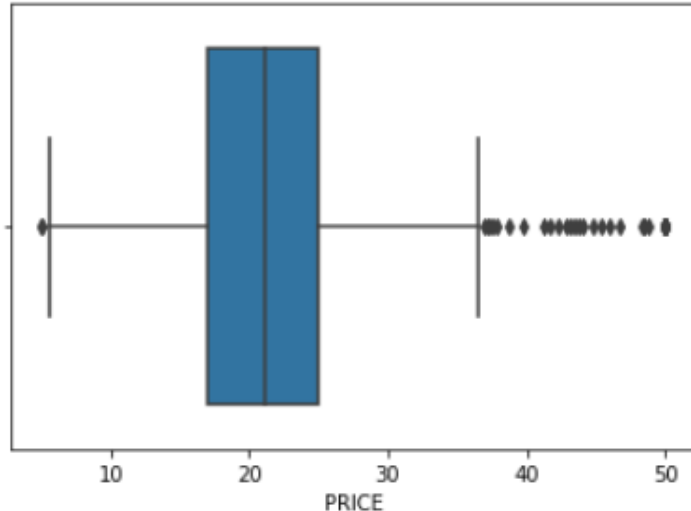
```
<matplotlib.axes._subplots.AxesSubplot at 0x7f44d082c670>
```



```
#Distribution using box plot
```

```
sns.boxplot(data.PRICE)
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f44d077ed60>
```



```
#Checking the correlation of the independent feature with the dependent feature
# Correlation is a statistical technique that can show whether and how strongly
pairs of variables are related. An intelligent correlation analysis can lead to a
greater understanding of your data
```

```
#checking Correlation of the data
```

```
correlation = data.corr()
```

```
correlation.loc['PRICE']
```

```
CRIM      -0.388305
```



```

ZN          0.360445
INDUS      -0.483725
CHAS       0.175260
NOX        -0.427321
RM         0.695360
AGE        -0.376955
DIS        0.249929
RAD        -0.381626
TAX        -0.468536
PTRATIO    -0.507787
B          0.333461
LSTAT      -0.737663
PRICE      1.000000

```

```
Name: PRICE, dtype: float64
```

```
# plotting the heatmap
```

```
import matplotlib.pyplot as plt
```

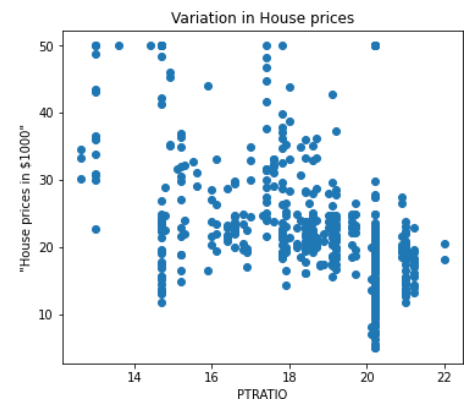
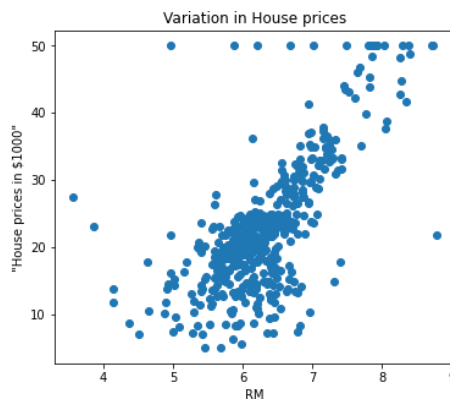
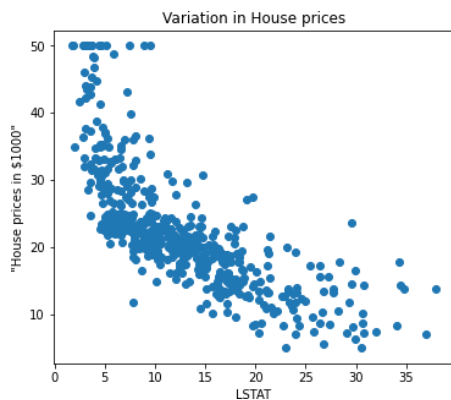
```
fig, axes = plt.subplots(figsize=(15,12))
```

```
sns.heatmap(correlation, square = True, annot = True)
```

```
# By looking at the correlation plot LSTAT is negatively correlated with -0.75 and
RM is positively correlated to the price and PTRATIO is correlated negatively
with -0.51
```



```
# Checking the scatter plot with the most correlated features
plt.figure(figsize = (20,5))
features = ['LSTAT','RM','PTRATIO']
for i, col in enumerate(features):
    plt.subplot(1, len(features) , i+1)
    x = data[col]
    y = data.PRICE
    plt.scatter(x, y, marker='o')
    plt.title("Variation in House prices")
    plt.xlabel(col)
    plt.ylabel('"House prices in $1000"')
```



```
# Splitting the dependent feature and independent feature
#X = data[['LSTAT','RM','PTRATIO']]
X = data.iloc[:, :-1]
y= data.PRICE
```

```
# In order to provide a standardized input to our neural network, we need the
perform the normalization of our dataset.
# This can be seen as an step to reduce the differences in scale that may arise
from the existent features.
# We perform this normalization by subtracting the mean from our data and
dividing it by the standard deviation.
# One more time, this normalization should only be performed by using the mean
and standard deviation from the training set,
# in order to avoid any information leak from the test set.
```

```
mean = X_train.mean(axis=0)
std = X_train.std(axis=0)
```

```
X_train = (X_train - mean) / std
X_test = (X_test - mean) / std
```

```
#Linear Regression
```

```
from sklearn.linear_model import LinearRegression
```

```
regressor = LinearRegression()
#Fitting the model
regressor.fit(X_train,y_train)
# Model Evaluation

#Prediction on the test dataset
y_pred = regressor.predict(X_test)
# Predicting RMSE the Test set results
from sklearn.metrics import mean_squared_error
rmse = (np.sqrt(mean_squared_error(y_test, y_pred)))
print(rmse)

from sklearn.metrics import r2_score
r2 = r2_score(y_test, y_pred)
print(r2)
# Neural Networks
#Scaling the dataset
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
# Due to the small amount of presented data in this dataset, we must be careful
to not create an overly complex model,
# which could lead to overfitting our data. For this, we are going to adopt an
architecture based on two Dense layers,
# the first with 128 and the second with 64 neurons, both using a ReLU activation
function.
# A dense layer with a linear activation will be used as output layer.
# In order to allow us to know if our model is properly learning, we will use a
mean squared error loss function and to report the performance of it we will
adopt the mean average error metric.
# By using the summary method from Keras, we can see that we have a total of
10,113 parameters, which is acceptable for us.

#Creating the neural network model
import keras
from keras.layers import Dense, Activation,Dropout
from keras.models import Sequential

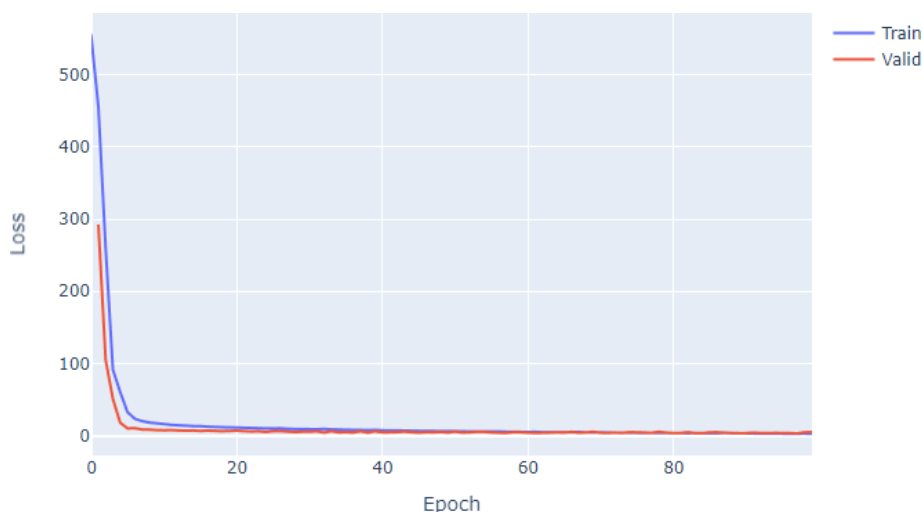
model = Sequential()
model.add(Dense(128,activation = 'relu',input_dim =13))
model.add(Dense(64,activation = 'relu'))
```

```
model.add(Dense(32,activation = 'relu'))
model.add(Dense(16,activation = 'relu'))
model.add(Dense(1))
#model.compile(optimizer='adam', loss='mse', metrics=['mae'])
model.compile(optimizer = 'adam',loss = 'mean_squared_error',metrics=['mae'])
!pip install ann_visualizer
!pip install graphviz
from ann_visualizer.visualize import ann_viz;
#Build your model here
ann_viz(model, title="DEMO ANN");
history = model.fit(X_train, y_train, epochs=100, validation_split=0.05)
# By plotting both loss and mean average error, we can see that our model was
capable of learning patterns in our data without overfitting taking place (as
shown by the validation set curves)
from plotly.subplots import make_subplots
import plotly.graph_objects as go

fig = go.Figure()
fig.add_trace(go.Scattergl(y=history.history['loss'],
                           name='Train'))

fig.add_trace(go.Scattergl(y=history.history['val_loss'],
                           name='Valid'))

fig.update_layout(height=500, width=700,
                  xaxis_title='Epoch',
                  yaxis_title='Loss')
fig.show()
```



```
fig = go.Figure()
```

```

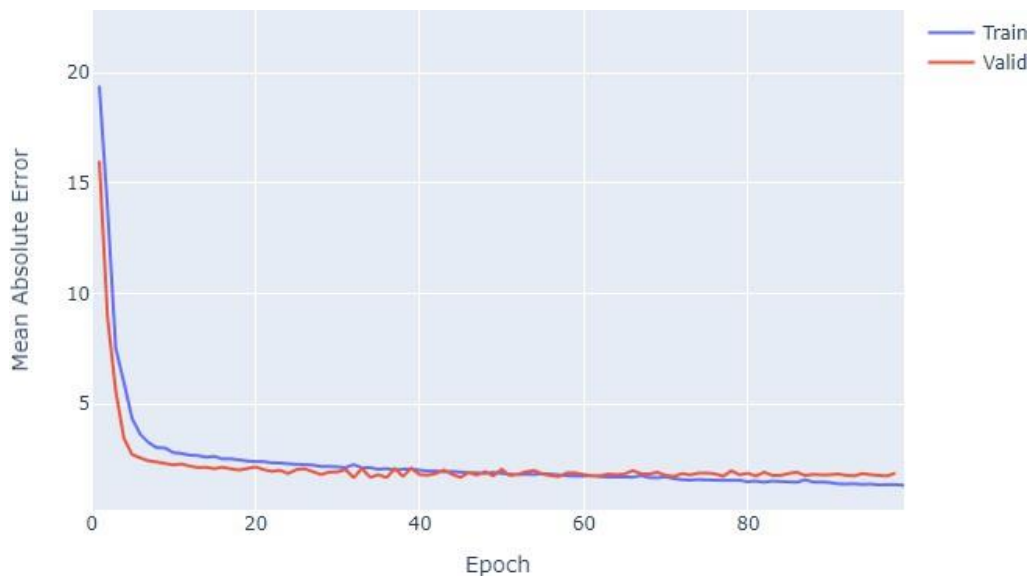
fig.add_trace(go.Scattergl(y=history.history['mae'],
                           name='Train'))

fig.add_trace(go.Scattergl(y=history.history['val_mae'],
                           name='Valid'))

fig.update_layout(height=500, width=700,
                  axis_title='Epoch',
                  yaxis_title='Mean Absolute Error')

fig.show()

```



```

#Evaluation of the model
y_pred = model.predict(X_test)
mse_nn, mae_nn = model.evaluate(X_test, y_test)

print('Mean squared error on test data: ', mse_nn)
print('Mean absolute error on test data: ', mae_nn)
4/4 [=====] - 0s 4ms/step - loss: 10.5717 - mae: 2.2670
Mean squared error on test data: 10.571733474731445 Mean
absolute error on test data: 2.2669904232025146
#Comparison with traditional approaches
#First let's try with a simple algorithm, the Linear Regression:
from sklearn.metrics import mean_absolute_error

lr_model = LinearRegression()
lr_model.fit(X_train, y_train)

```

```

y_pred_lr = lr_model.predict(X_test)
mse_lr = mean_squared_error(y_test, y_pred_lr)
mae_lr = mean_absolute_error(y_test, y_pred_lr)

print('Mean squared error on test data: ', mse_lr)
print('Mean absolute error on test data: ', mae_lr)
from sklearn.metrics import r2_score
r2 = r2_score(y_test, y_pred)
print(r2)
0.8812832788381159

# Predicting RMSE the Test set results
from sklearn.metrics import mean_squared_error
rmse = (np.sqrt(mean_squared_error(y_test, y_pred)))
print(rmse)
3.320768607496587

# Make predictions on new data
import sklearn
new_data = sklearn.preprocessing.StandardScaler().fit_transform([[0.1, 10.0,
5.0, 0, 0.4, 6.0, 50, 6.0, 1, 400, 20, 300, 10]])
prediction = model.predict(new_data)
print("Predicted house price:", prediction)
1/1 [=====] - 0s 70ms/step
Predicted house price: [[11.104753]]

```

#new_data = sklearn.preprocessing.StandardScaler().fit_transform([[0.1, 10.0, 5.0, 0, 0.4, 6.0, 50, 6.0, 1, 400, 20, 300, 10]]) is a line of code that standardizes the input features of a new data point.

In this specific case, we have a new data point represented as a list of 13 numeric values ([0.1, 10.0, 5.0, 0, 0.4, 6.0, 50, 6.0, 1, 400, 20, 300, 10]) that represents the values for the 13 features of the Boston House Price dataset.

The StandardScaler() function from the sklearn.preprocessing module is used to standardize the data. Standardization scales each feature to have zero mean and unit variance, which is a common preprocessing step in machine learning to ensure that all features contribute equally to the model.

The fit_transform() method is used to fit the scaler to the data and apply the standardization transformation. The result is a new data

point with standardized feature values.

Conclusion- In this way we can Predict the Boston House Price using Deep Neural Network.

Group B Deep Learning

Assignment No: 3

Title of the Assignment: Use MNIST Fashion Dataset and create a classifier to classify fashion clothing into categories.

Objective of the Assignment: Students should be able to Classify movie reviews into positive reviews and "negative reviews on IMDB Dataset.

Prerequisite:

1. Basic of programming language
 2. Concept of Classification
 3. Concept of Deep Neural Network
-

Contents for Theory:

1. What is Classification
 2. Example of Classification
 3. What is CNN?
 4. How Deep Neural Network Work on Classification
 5. Code Explanation with Output
-

What is Classification?

Classification is a type of supervised learning in machine learning that involves categorizing data into predefined classes or categories based on a set of features or characteristics. It is used to predict the class of new, unseen data based on the patterns learned from the labeled training data.

In classification, a model is trained on a labeled dataset, where each data point has a known class label. The model learns to associate the input features with the corresponding class labels and can then be used to classify new, unseen data.

For example, we can use classification to identify whether an email is spam or not based on its content and metadata, to predict whether a patient has a disease based on their medical records and symptoms, or to classify images into different categories based on their visual features.

Classification algorithms can vary in complexity, ranging from simple models such as decision trees and k-nearest neighbors to more complex models such as support vector machines and neural networks. The choice of algorithm depends on the nature of the data, the size of the dataset, and the desired level of accuracy and interpretability.

Example- Classification is a common task in deep neural networks, where the goal is to predict the class of an input based on its features. Here's an example of how classification can be performed in a deep neural network using the popular MNIST dataset of handwritten digits.

The MNIST dataset contains 60,000 training images and 10,000 testing images of handwritten digits from 0 to 9. Each image is a grayscale 28x28 pixel image, and the task is to classify each image into one of the 10 classes corresponding to the 10 digits.

We can use a convolutional neural network (CNN) to classify the MNIST dataset. A CNN is a type of deep neural network that is commonly used for image classification tasks.

What is CNN-

Convolutional Neural Networks (CNNs) are commonly used for image classification tasks, and they are designed to automatically learn and extract features from input images. Let's consider an example of using a CNN to classify images of handwritten digits.

In a typical CNN architecture for image classification, there are several layers, including convolutional layers, pooling layers, and fully connected layers. Here's a diagram of a simple CNN architecture for the digit classification task:

The input to the network is an image of size 28x28 pixels, and the output is a probability distribution over the 10 possible digits (0 to 9).

The convolutional layers in the CNN apply filters to the input image, looking for specific patterns and features. Each filter produces a feature map that highlights areas of the image that match the filter. The filters are learned during training, so the network can automatically learn which features are most relevant for the classification task.

The pooling layers in the CNN downsample the feature maps, reducing the spatial dimensions of the data. This helps to reduce the number of parameters in the network, while also making the features more robust to small variations in the input image.

The fully connected layers in the CNN take the flattened output from the last pooling layer and perform a classification task by outputting a probability distribution over the 10 possible digits.

During training, the network learns the optimal values of the filters and parameters by minimizing a loss function. This is typically done using stochastic gradient descent or a similar optimization algorithm.

Once trained, the network can be used to classify new images by passing them through the network and computing the output probability distribution.

Overall, CNNs are powerful tools for image recognition tasks and have been used successfully in many applications, including object detection, face recognition, and medical image analysis.

CNNs have a wide range of applications in various fields, some of which are:

Image classification: CNNs are commonly used for image classification tasks, such as identifying objects in images and recognizing faces.

Object detection: CNNs can be used for object detection in images and videos, which involves identifying the location of objects in an image and drawing bounding boxes around them.

Semantic segmentation: CNNs can be used for semantic segmentation, which involves partitioning an image into segments and assigning each segment a semantic label (e.g., "road", "sky", "building").

Natural language processing: CNNs can be used for natural language processing tasks, such as sentiment analysis and text classification.

Medical imaging: CNNs are used in medical imaging for tasks such as diagnosing diseases from X-rays and identifying tumors from MRI scans.

Autonomous vehicles: CNNs are used in autonomous vehicles for tasks such as object detection and lane detection.

Video analysis: CNNs can be used for tasks such as video classification, action recognition, and video captioning.

Overall, CNNs are a powerful tool for a wide range of applications, and they have been used successfully in many areas of research and industry.

How Deep Neural Network Work on Classification using CNN-

Deep neural networks using CNNs work on classification tasks by learning to automatically extract features from input images and using those features to make predictions. Here's how it works:

Input layer: The input layer of the network takes in the image data as input.

Convolutional layers: The convolutional layers apply filters to the input images to extract relevant features. Each filter produces a feature map that highlights areas of the image that match the filter.

Activation functions: An activation function is applied to the output of each convolutional layer to introduce non-linearity into the network.

Pooling layers: The pooling layers downsample the feature maps to reduce the spatial dimensions of the data.

Dropout layer: Dropout is used to prevent overfitting by randomly dropping out a percentage of the neurons in the network during training.

Fully connected layers: The fully connected layers take the flattened output from the last pooling layer and perform a classification task by outputting a probability distribution over the possible classes.

Softmax activation function: The softmax activation function is applied to the output of the last fully connected layer to produce a probability distribution over the possible classes.

Loss function: A loss function is used to compute the difference between the predicted probabilities and the actual labels.

Optimization: An optimization algorithm, such as stochastic gradient descent, is used to minimize the loss function by adjusting the values of the network parameters.

Training: The network is trained on a large dataset of labeled images, adjusting the values of the parameters to minimize the loss function.

Prediction: Once trained, the network can be used to classify new images by passing them through the network and computing the output probability distribution.

MNIST Dataset-

The MNIST Fashion dataset is a collection of 70,000 grayscale images of 28x28 pixels, representing 10 different categories of clothing and accessories. The categories include T-shirts/tops, trousers, pullovers, dresses, coats, sandals, shirts, sneakers, bags, and ankle boots.

The dataset is often used as a benchmark for testing image classification algorithms, and it is considered a more challenging version of the original MNIST dataset which contains handwritten digits. The

MNIST Fashion dataset was released by Zalando Research in 2017 and has since become a popular dataset in the machine learning community.

The MNIST Fashion dataset is a collection of 70,000 grayscale images of 28x28 pixels each. These images represent 10 different categories of clothing and accessories, with each category containing 7,000 images. The categories are as follows:

T-shirt/tops

Trousers

Pullovers

Dresses

Coats

Sandals

Shirts

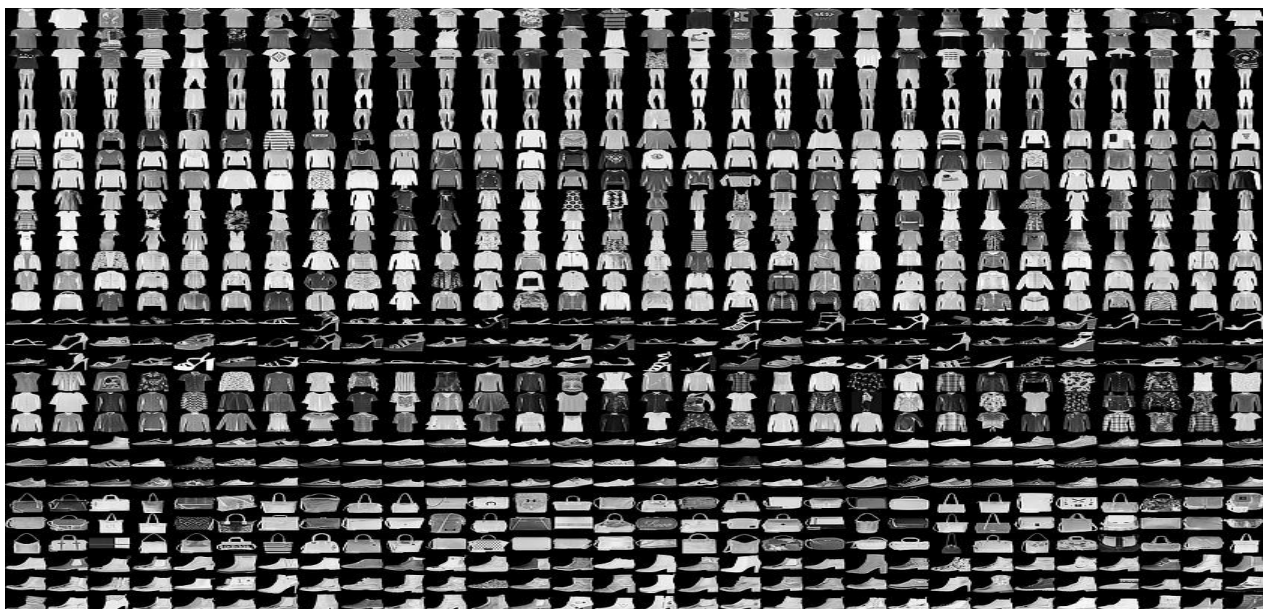
Sneakers

Bags

Ankle boots

The images were obtained from Zalando's online store and are preprocessed to be normalized and centered. The training set contains 60,000 images, while the test set contains 10,000 images. The goal of the dataset is to accurately classify the images into their respective categories.

The MNIST Fashion dataset is often used as a benchmark for testing image classification algorithms, and it is considered a more challenging version of the original MNIST dataset which contains handwritten digits. The dataset is widely used in the machine learning community for research and educational purposes.



Here are the general steps to perform Convolutional Neural Network (CNN) on the MNIST Fashion dataset:

- Import the necessary libraries, including TensorFlow, Keras, NumPy, and Matplotlib.
- Load the dataset using Keras' built-in function, `keras.datasets.fashion_mnist.load_data()`. This will provide the training and testing sets, which will be used to train and evaluate the CNN.
- Preprocess the data by normalizing the pixel values between 0 and 1, and reshaping the images to be of size (28, 28, 1) for compatibility with the CNN.
- Define the CNN architecture, including the number and size of filters, activation functions, and pooling layers. This can vary based on the specific problem being addressed.
- Compile the model by specifying the loss function, optimizer, and evaluation metrics. Common choices include categorical cross-entropy, Adam optimizer, and accuracy metric.
- Train the CNN on the training set using the `fit()` function, specifying the number of epochs and batch size.
- Evaluate the performance of the model on the testing set using the `evaluate()` function. This will provide metrics such as accuracy and loss on the test set.
- Use the trained model to make predictions on new images, if desired, using the `predict()` function.

Source Code with Output-

```
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow import keras
import numpy as np
```

```
(x_train, y_train), (x_test, y_test) = keras.datasets.fashion_mnist.load_data()
```

There are 10 image classes in this dataset and each class has a mapping corresponding to the following labels:

```
#0 T-shirt/top
#1 Trouser
#2 pullover
#3 Dress
#4 Coat
#5 sandals
```

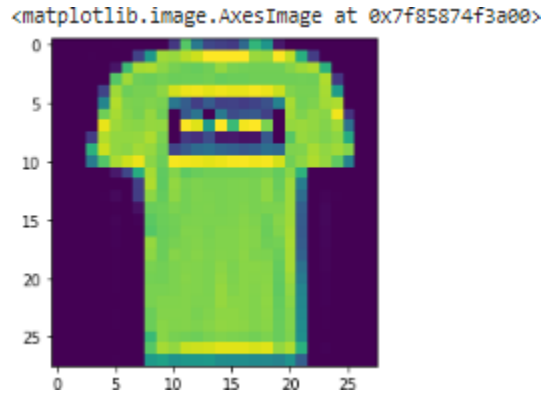
#6 shirt

#7 sneaker

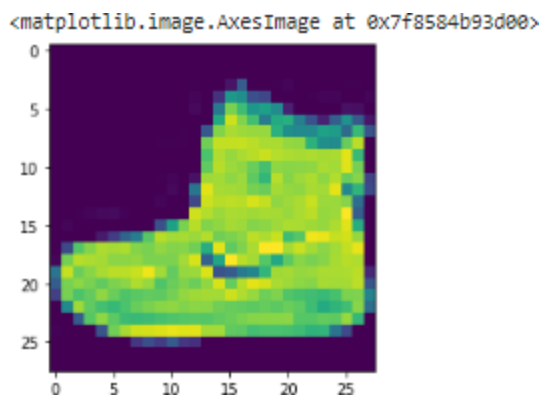
#8 bag

#9 ankle boot

```
plt.imshow(x_train[1])
```



```
plt.imshow(x_train[0])
```



Next, we will preprocess the data by scaling the pixel values to be between 0 and 1, and then reshaping the images to be 28x28 pixels.

```
x_train = x_train.astype('float32') / 255.0
```

```
x_test = x_test.astype('float32') / 255.0
```

```
x_train = x_train.reshape(-1, 28, 28, 1)
```

```
x_test = x_test.reshape(-1, 28, 28, 1)
```

28, 28 comes from width, height, 1 comes from the number of channels

-1 means that the length in that dimension is inferred.

This is done based on the constraint that the number of elements in an ndarray or Tensor when reshaped must remain the same.

```
# each image is a row vector (784 elements) and there are lots of such rows (let it be n, so there are 784n
elements). So TensorFlow can infer that -1 is n.
# converting the training_images array to 4 dimensional array with sizes 60000, 28, 28, 1 for 0th to 3rd
dimension.
x_train.shape
(60000, 28, 28)
x_test.shape
(10000, 28, 28, 1)
y_train.shape
(60000,)
y_test.shape
(10000,)
# We will use a convolutional neural network (CNN) to classify the fashion items.
# The CNN will consist of multiple convolutional layers followed by max pooling,
# dropout, and dense layers. Here is the code for the model:
model = keras.Sequential([
    keras.layers.Conv2D(32, (3,3), activation='relu', input_shape=(28,28,1)),
    # 32 filters (default), randomly initialized
    # 3*3 is Size of Filter
    # 28,28,1 size of Input Image
    # No zero-padding: every output 2 pixels less in every dimension
    # in Paramter shwon 320 is value of weights: (3x3 filter weights + 32 bias) * 32 filters
    #  $32*3*3=288(\text{Total})+32(\text{bias})=320$ 
    keras.layers.MaxPooling2D((2,2)),
    # It shown 13 * 13 size image with 32 channel or filter or depth.
    keras.layers.Dropout(0.25),
    # Reduce Overfitting of Training sample drop out 25% Neuron
    keras.layers.Conv2D(64, (3,3), activation='relu'),
    # Deeper layers use 64 filters
    # 3*3 is Size of Filter
    # Observe how the input image on 28x28x1 is transformed to a 3x3x64 feature map
    #  $13(\text{Size})-3(\text{Filter Size})+1(\text{bias})=11$  Size for Width and Height with 64 Depth or fltter or channel
    # in Paramter shwon 18496 is value of weights: (3x3 filter weights + 64 bias) * 64 filters
    #  $64*3*3=576+1=577*32 + 32(\text{bias})=18496$ 
    keras.layers.MaxPooling2D((2,2)),
    # It shown 5 * 5 size image with 64 channel or filter or depth.
    keras.layers.Dropout(0.25),
```

```

keras.layers.Conv2D(128, (3,3), activation='relu'),
# Deeper layers use 128 filters
# 3*3 is Size of Filter
# Observe how the input image on 28x28x1 is transformed to a 3x3x128 feature map
# It show 5(Size)-3(Filter Size )+1(bias)=3 Size for Width and Height with 64 Depth or filter or
channel
# 128*3*3=1152+1=1153*64 + 64(bias)= 73856
# To classify the images, we still need a Dense and Softmax layer.
# We need to flatten the 3x3x128 feature map to a vector of size 1152
keras.layers.Flatten(),
keras.layers.Dense(128, activation='relu'),
# 128 Size of Node in Dense Layer
# 1152*128 = 147584
keras.layers.Dropout(0.25),
keras.layers.Dense(10, activation='softmax')
# 10 Size of Node another Dense Layer
# 128*10+10 bias= 1290
])
model.summary()
Model: "sequential"

```

Layer (type)	Output Shape	Param #
=====		
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
)		
dropout (Dropout)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 11, 11, 64)	18496
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
2D)		
dropout_1 (Dropout)	(None, 5, 5, 64)	0

conv2d_2 (Conv2D)	(None, 3, 3, 128)	73856
flatten (Flatten)	(None, 1152)	0
dense (Dense)	(None, 128)	147584
dropout_2 (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1290

=====

Total params: 241,546

Trainable params: 241,546

Non-trainable params: 0

Compile and Train the Model

After defining the model, we will compile it and train it on the training data.

```
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])
```

```
history = model.fit(x_train, y_train, epochs=10, validation_data=(x_test, y_test))
```

1875 is a number of batches. By default batches contain 32 samples. $60000 / 32 = 1875$

Finally, we will evaluate the performance of the model on the test data.

```
test_loss, test_acc = model.evaluate(x_test, y_test)
```

```
print("Test accuracy:", test_acc)
```

```
313/313 [=====] - 3s 10ms/step - loss: 0.2606 - accuracy: 0.9031
```

Test accuracy: 0.9031000137329102

Conclusion- In this way we can Classify fashion clothing into categories using CNN.

Group B Deep Learning

Assignment No: 2

Title of the Assignment: Binary classification using Deep Neural Networks Example: Classify movie reviews into positive" reviews and "negative" reviews, just based on the text content of the reviews. Use IMDB dataset

Objective of the Assignment: Students should be able to Classify movie reviews into positive reviews and "negative reviews on IMDB Dataset.

Prerequisite:

1. Basic of programming language
 2. Concept of Classification
 3. Concept of Deep Neural Network
-

Contents for Theory:

1. What is Classification
 2. Example of Classification
 3. How Deep Neural Network Work on Classification
 4. Code Explanation with Output
-

What is Classification?

Classification is a type of supervised learning in machine learning that involves categorizing data into predefined classes or categories based on a set of features or characteristics. It is used to predict the class of new, unseen data based on the patterns learned from the labeled training data.

In classification, a model is trained on a labeled dataset, where each data point has a known class label. The model learns to associate the input features with the corresponding class labels and can then be used to classify new, unseen data.

For example, we can use classification to identify whether an email is spam or not based on its content and metadata, to predict whether a patient has a disease based on their medical records and symptoms, or to classify images into different categories based on their visual features.

Classification algorithms can vary in complexity, ranging from simple models such as decision trees and k-nearest neighbors to more complex models such as support vector machines and neural networks. The choice of algorithm depends on the nature of the data, the size of the dataset, and the desired level of accuracy and interpretability.

Classification is a common task in deep neural networks, where the goal is to predict the class of an input based on its features. Here's an example of how classification can be performed in a deep neural network using the popular MNIST dataset of handwritten digits.

The MNIST dataset contains 60,000 training images and 10,000 testing images of handwritten digits from 0 to 9. Each image is a grayscale 28x28 pixel image, and the task is to classify each image into one of the 10 classes corresponding to the 10 digits.

We can use a convolutional neural network (CNN) to classify the MNIST dataset. A CNN is a type of deep neural network that is commonly used for image classification tasks.

How Deep Neural Network Work on Classification-

Deep neural networks are commonly used for classification tasks because they can automatically learn to extract relevant features from raw input data and map them to the correct output class.

The basic architecture of a deep neural network for classification consists of three main parts: an input layer, one or more hidden layers, and an output layer. The input layer receives the raw input data, which is usually preprocessed to a fixed size and format. The hidden layers are composed of neurons that apply linear transformations and nonlinear activations to the input features to extract relevant patterns and representations. Finally, the output layer produces the predicted class labels, usually as a probability distribution over the possible classes.

During training, the deep neural network learns to adjust its weights and biases in each layer to minimize the difference between the predicted output and the true labels. This is typically done by optimizing a loss function that measures the discrepancy between the predicted and true labels, using techniques such as gradient descent or stochastic gradient descent.

One of the key advantages of deep neural networks for classification is their ability to learn hierarchical representations of the input data. In a deep neural network with multiple hidden layers, each layer learns to capture more complex and abstract features than the previous layer, by building on the representations learned by the earlier layers. This hierarchical structure allows deep neural networks to learn highly discriminative features that can separate different classes of input data, even when the data is highly complex or noisy.

Overall, the effectiveness of deep neural networks for classification depends on the choice of architecture, hyperparameters, and training procedure, as well as the quality and quantity of the training data. When trained properly, deep neural networks can achieve state-of-the-art performance on a wide range of classification tasks, from image recognition to natural language processing.

IMDB Dataset-The IMDB dataset is a large collection of movie reviews collected from the IMDB website, which is a popular source of user-generated movie ratings and reviews. The dataset consists of 50,000 movie reviews, split into 25,000 reviews for training and 25,000 reviews for testing.

Each review is represented as a sequence of words, where each word is represented by an integer index based on its frequency in the dataset. The labels for each review are binary, with 0 indicating a negative review and 1 indicating a positive review.

The IMDB dataset is commonly used as a benchmark for sentiment analysis and text classification tasks, where the goal is to classify the movie reviews as either positive or negative based on their text content. The dataset is challenging because the reviews are often highly subjective and can contain complex language and nuances of meaning, making it difficult for traditional machine learning approaches to accurately classify them.

Deep learning approaches, such as deep neural networks, have achieved state-of-the-art performance on the IMDB dataset by automatically learning to extract relevant features from the raw text data and map them to the correct output class. The IMDB dataset is widely used in research and education for natural language processing and machine learning, as it provides a rich source of labeled text data for training and testing deep learning models.

Source Code and Output-

The IMDB sentiment classification dataset consists of 50,000 movie reviews from IMDB users that are labeled as either positive (1) or negative (0).

The reviews are preprocessed and each one is encoded as a sequence of word indexes in the form of integers.

The words within the reviews are indexed by their overall frequency within the dataset. For example, the integer "2" encodes the second most frequent word in the data.

The 50,000 reviews are split into 25,000 for training and 25,000 for testing.

Text Process word by word at different timestamp (You may use RNN LSTM GRU)

convert input text to vector represent input text

DOMAIN: Digital content and entertainment industry

CONTEXT: The objective of this project is to build a text classification model that analyses the customer's sentiments based on their reviews in the IMDB database. The model uses a complex deep learning model to build an embedding layer followed by a classification algorithm to analyse the sentiment of the customers.

DATA DESCRIPTION: The Dataset of 50,000 movie reviews from IMDB, labelled by sentiment (positive/negative).

Reviews have been preprocessed, and each review is encoded as a sequence of word indexes (integers).

For convenience, the words are indexed by their frequency in the dataset, meaning the word that has index 1 is the most frequent word.

Use the first 20 words from each review to speed up training, using a max vocabulary size of 10,000.

As a convention, "0" does not stand for a specific word, but instead is used to encode any unknown word.

PROJECT OBJECTIVE: Build a sequential NLP classifier which can use input text parameters to determine the customer sentiments.

```
import numpy as np
```

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
#loading imdb data with most frequent 10000 words
```

```
from keras.datasets import imdb
(X_train, y_train), (X_test, y_test) = imdb.load_data(num_words=10000) # you may take top 10,000
word frequently used review of movies other are discarded
#consolidating data for EDA Exploratory data analysis (EDA) is used by data scientists to analyze and
investigate data sets and summarize their main characteristics
data = np.concatenate((X_train, X_test), axis=0) # axis 0 is first running vertically downwards across
rows (axis 0), axis 1 is second running horizontally across columns (axis 1),
label = np.concatenate((y_train, y_test), axis=0)
X_train.shape
(25000,)
X_test.shape
(25000,)
y_train.shape
(25000,)
y_test.shape
(25000,)
print("Review is ",X_train[0]) # series of no converted word to vocabulary associated with index
print("Review is ",y_train[0])
Review is [1, 194, 1153, 194, 8255, 78, 228, 5, 6, 1463, 4369, 5012, 134, 26, 4, 715, 8, 118, 1634, 14,
394, 20, 13, 119, 954, 189, 102, 5, 207, 110, 3103, 21, 14, 69, 188, 8, 30, 23, 7, 4, 249, 126, 93, 4, 114,
9, 2300, 1523, 5, 647, 4, 116, 9, 35, 8163, 4, 229, 9, 340, 1322, 4, 118, 9, 4, 130, 4901, 19, 4, 1002, 5,
89, 29, 952, 46, 37, 4, 455, 9, 45, 43, 38, 1543, 1905, 398, 4, 1649, 26, 6853, 5, 163, 11, 3215, 2, 4,
1153, 9, 194, 775, 7, 8255, 2, 349, 2637, 148, 605, 2, 8003, 15, 123, 125, 68, 2, 6853, 15, 349, 165,
4362, 98, 5, 4, 228, 9, 43, 2, 1157, 15, 299, 120, 5, 120, 174, 11, 220, 175, 136, 50, 9, 4373, 228, 8255,
5, 2, 656, 245, 2350, 5, 4, 9837, 131, 152, 491, 18, 2, 32, 7464, 1212, 14, 9, 6, 371, 78, 22, 625, 64,
1382, 9, 8, 168, 145, 23, 4, 1690, 15, 16, 4, 1355, 5, 28, 6, 52, 154, 462, 33, 89, 78, 285, 16, 145, 95]
Review is 0
vocab=imdb.get_word_index() # Retrieve the word index file mapping words to indices
print(vocab)
{'fawn': 34701, 'tsukino': 52006, 'nunnery': 52007, 'sonja': 16816, 'vani': 63951, 'woods': 1408, 'spiders':
16115,
y_train
array([1, 0, 0, ..., 0, 1, 0])
y_test
array([0, 1, 1, ..., 0, 0, 0])
# Function to perform relevant sequence adding on the data
# Now it is time to prepare our data. We will vectorize every review and fill it with zeros so that it
```

contains exactly 10000 numbers.

That means we fill every review that is shorter than 500 with zeros.

We do this because the biggest review is nearly that long and every input for our neural network needs to have the same size.

We also transform the targets into floats.

sequences is name of method the review less than 10000 we perform padding overthere

binary vectorization code:

VECTORIZE as one cannot feed integers into a NN

Encoding the integer sequences into a binary matrix - one hot encoder basically

From integers representing words, at various lengths - to a normalized one hot encoded tensor (matrix) of 10k columns

`def vectorize(sequences, dimension = 10000):` # We will vectorize every review and fill it with zeros so that it contains exactly 10,000 numbers.

 # Create an all-zero matrix of shape (len(sequences), dimension)

`results = np.zeros((len(sequences), dimension))`

`for i, sequence in enumerate(sequences):`

`results[i, sequence] = 1`

`return results`

Now we split our data into a training and a testing set.

The training set will contain reviews and the testing set

Set a VALIDATION set

`test_x = data[:10000]`

`test_y = label[:10000]`

`train_x = data[10000:]`

`train_y = label[10000:]`

`test_x.shape`

`(10000,)`

`test_y.shape`

`(10000,)`

`train_x.shape`

`(40000,)`

`train_y.shape`

`(40000,)`

`print("Categories:", np.unique(label))`

`print("Number of unique words:", len(np.unique(np.hstack(data))))`

The `hstack()` function is used to stack arrays in sequence horizontally (column wise).

Categories: [0 1]

Number of unique words: 9998

`length = [len(i) for i in data]`

`print("Average Review length:", np.mean(length))`

`print("Standard Deviation:", round(np.std(length)))`

The whole dataset contains 9998 unique words and the average review length is 234 words, with a standard deviation of 173 words.

Average Review length: 234.75892

Standard Deviation: 173

If you look at the data you will realize it has been already pre-processed.

All words have been mapped to integers and the integers represent the words sorted by their frequency.

This is very common in text analysis to represent a dataset like this.

So 4 represents the 4th most used word,

5 the 5th most used word and so on...

The integer 1 is reserved for the start marker,

the integer 2 for an unknown word and 0 for padding.

Let's look at a single training example:

`print("Label:", label[0])`

Label: 1

`print("Label:", label[1])`

Label: 0

`print(data[0])`

Retrieves a dict mapping words to their index in the IMDB dataset.

`index = imdb.get_word_index() # word to index`

Create inverted index from a dictionary with document ids as keys and a list of terms as values for each document

`reverse_index = dict([(value, key) for (key, value) in index.items()]) # id to word`

`decoded = " ".join([reverse_index.get(i - 3, "#") for i in data[0]])`

The indices are offset by 3 because 0, 1 and 2 are reserved indices for "padding", "start of sequence" and "unknown".

`print(decoded)`

this film was just brilliant casting location scenery story direction everyone's really suited the part they

played and you could just imagine being there robert # is an amazing actor and now the same being director # father came from the same scottish island as myself so i loved the fact there was a real connection with this film the witty remarks throughout the film

#Adding sequence to data

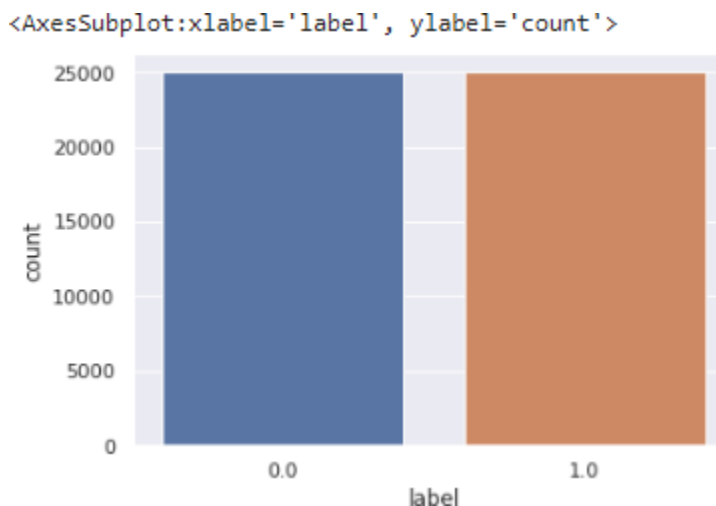
Vectorization is the process of converting textual data into numerical vectors and is a process that is usually applied once the text is cleaned.

```
data = vectorize(data)
```

```
label = np.array(label).astype("float32")
```

```
labelDF=pd.DataFrame({'label':label})
```

```
sns.countplot(x='label', data=labelDF)
```



Creating train and test data set

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(data,label, test_size=0.20, random_state=1)
```

```
X_train.shape
```

```
(40000, 10000)
```

```
X_test.shape
```

```
(10000, 10000)
```

Let's create sequential model

```
from keras.utils import to_categorical
```

```
from keras import models
```

```
from keras import layers
```

```
model = models.Sequential()
```

Input - Layer

Note that we set the input-shape to 10,000 at the input-layer because our reviews are 10,000 integers long.

The input-layer takes 10,000 as input and outputs it with a shape of 50.

```
model.add(layers.Dense(50, activation = "relu", input_shape=(10000, )))
```

Hidden - Layers

Please note you should always use a dropout rate between 20% and 50%. # here in our case 0.3 means 30% dropout we are using dropout to prevent overfitting.

By the way, if you want you can build a sentiment analysis without LSTMs, then you simply need to replace it by a flatten layer:

```
model.add(layers.Dropout(0.3, noise_shape=None, seed=None))
```

```
model.add(layers.Dense(50, activation = "relu"))
```

```
model.add(layers.Dropout(0.2, noise_shape=None, seed=None))
```

```
model.add(layers.Dense(50, activation = "relu"))
```

Output- Layer

```
model.add(layers.Dense(1, activation = "sigmoid"))
```

```
model.summary()
```

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
=====		
dense (Dense)	(None, 50)	500050
dropout (Dropout)	(None, 50)	0
dense_1 (Dense)	(None, 50)	2550
dropout_1 (Dropout)	(None, 50)	0
dense_2 (Dense)	(None, 50)	2550
dense_3 (Dense)	(None, 1)	51

```
=====
```

```
Total params: 505,201
```

```
Trainable params: 505,201
```

```
Non-trainable params: 0
```

```
#For early stopping
```

```
# Stop training when a monitored metric has stopped improving.
```

```
# monitor: Quantity to be monitored.
```

```
# patience: Number of epochs with no improvement after which training will be stopped.
```

```
import tensorflow as tf
```

```
callback = tf.keras.callbacks.EarlyStopping(monitor='loss', patience=3)
```

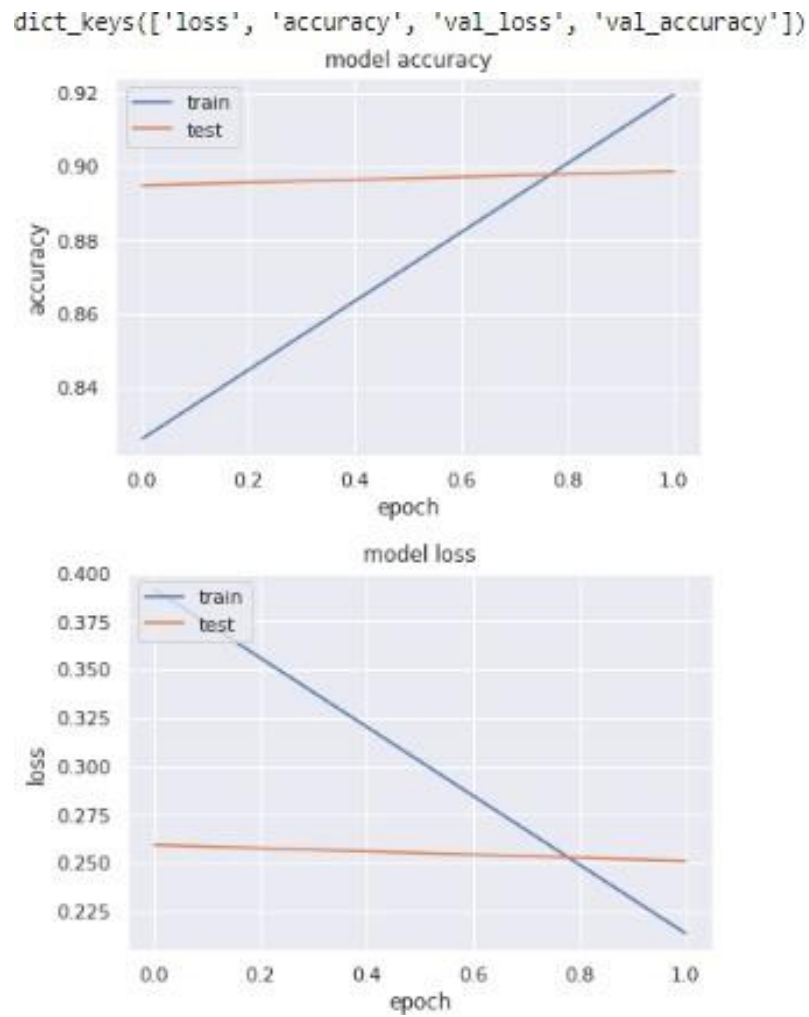
```
# We use the "adam" optimizer, an algorithm that changes the weights and biases during training.
```

```
# We also choose binary-crossentropy as loss (because we deal with binary
```

classification) and accuracy as our evaluation metric.

```
model.compile(  
    optimizer = "adam",  
    loss = "binary_crossentropy",  
    metrics = ["accuracy"]  
)  
from sklearn.model_selection import train_test_split  
  
results = model.fit(  
    X_train, y_train,  
    epochs= 2,  
    batch_size = 500,  
    validation_data = (X_test, y_test),  
    callbacks=[callback]  
)  
# Let's check mean accuracy of our model  
print(np.mean(results.history["val_accuracy"]))  
# Evaluate the model  
score = model.evaluate(X_test, y_test, batch_size=500)  
print('Test loss:', score[0])  
print('Test accuracy:', score[1])  
20/20 [=====] - 1s 24ms/step - loss: 0.2511 - accuracy:  
0.8986  
Test loss: 0.25108325481414795  
Test accuracy: 0.8985999822616577  
#Let's plot training history of our model.  
  
# list all data in history  
print(results.history.keys())  
# summarize history for accuracy  
plt.plot(results.history['accuracy'])  
plt.plot(results.history['val_accuracy'])  
plt.title('model accuracy')  
plt.ylabel('accuracy')  
plt.xlabel('epoch')  
plt.legend(['train', 'test'], loc='upper left')  
plt.show()  
# summarize history for loss  
plt.plot(results.history['loss'])  
plt.plot(results.history['val_loss'])  
plt.title('model loss')  
plt.ylabel('loss')
```

```
plt.xlabel('epoch')  
plt.legend(['train', 'test'], loc='upper left')  
plt.show()
```



Conclusion- In this way we can Classify the Movie Reviews by using DNN.